Inertia and balanced forces

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Predict motion when the net force is zero.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 6.*
3 m east and 4 m south. Add each pair of signed components.

02. Worked example: explain the model
Equal opposite forces; any horizontal v-t line, including v=0, is consistent.

03. Balanced while moving
8-8=0 N. It continues right at the same velocity.

04. After release
Hand force becomes zero; puck keeps constant horizontal velocity. Weight and support balance.

05. The bus stops
Passenger initially shares forward velocity and tends to continue. Seatbelt/contact slows passenger; no extra forward force is required.

06. Review: connect an earlier idea *Review: Unit 1, Day 8.*
a=3.333 m/s² east; average velocity 5 m/s east; displacement 15 m east.

07. Exit check
First two can. Turning and speeding up change velocity and require nonzero net force.

08. STRETCH: test the model
Upward support balances weight; no horizontal force. Existing rightward velocity persists.

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01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 6.*
R=(2, 4) m; magnitude ≈4.472 m at 63.43° north of east. C=(-2, -4) m.

02. Worked example: explain the model
Equal opposite forces; any horizontal v-t line, including v=0, is consistent.

03. Infer the motion constraint
a=0; velocity remains -3 m/s;Δx=-15 m.

04. Inertia in a braking bus
Passenger keeps initial velocity until belt/seat/contact provides braking force. Bus frame accelerates; apparent forward drift does not require a new forward interaction.

05. Engine-off spacecraft
Δr=(40,-30) km; magnitude 50 km; constant velocity without net force.

06. Review: connect an earlier idea *Review: Unit 1, Day 8.*
Stops after 5 s at distance 37.5 m. Extending to 7 s predicts -6 m/s, which assumes continued acceleration after stopping and reversal.

07. Exit check
Direction changes, so velocity changes and acceleration/net force are nonzero, inward for uniform circular motion.

08. STRETCH: test the model
Constant-velocity tension 5000 N. Tension below 5000 N gives downward acceleration, so an upward-moving elevator slows initially.